

APZ lab3 report new

Docker Compose Status: A terminal view showing the docker compose ps output. It verifies that all required containers are healthy and running, specifically listing the PostgreSQL database (db), three Hazelcast nodes (hz1, hz2, hz3), three logging services (ls1, ls2, ls3), the counter service (cs), and the facade service

```
(.venv_APZ_general) julfy@julfy-Nitro-ANS15-57:~/Documents/6th_term/APZ/Task_3_micro_hazel/APZ_labs$ docker compose up -d --remove-orphans --wait
[+] Running 10/10
✓ Container config_server Removed
✓ Container hz3 Healthy
✓ Container db Healthy
✓ Container hz1 Healthy
✓ Container hz2 Healthy
✓ Container cs Healthy
✓ Container ls2 Healthy
✓ Container ls3 Healthy
✓ Container ls1 Healthy
✓ Container fs Healthy
(.venv_APZ_general) julfy@julfy-Nitro-ANS15-57:~/Documents/6th_term/APZ/Task_3_micro_hazel/APZ_labs$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
cs	apz_labs-counter-service	"uvicorn main:app --..."	counter-service	2 minutes ago	Up About a minute	8000/tcp
db	postgres:15	"docker-entrypoint.s..."	postgres	2 minutes ago	Up 2 minutes (healthy)	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
fs	apz_labs-facade-service	"uvicorn main:app --..."	facade-service	2 minutes ago	Up 27 seconds	
hz1	hazelcast/hazelcast:5.3	"hz start"	hazelcast1	2 minutes ago	Up 2 minutes	0.0.0.0:5701->5701/tcp, [::]:5701->5701/tcp
hz2	hazelcast/hazelcast:5.3	"hz start"	hazelcast2	2 minutes ago	Up 2 minutes	0.0.0.0:5702->5701/tcp, [::]:5702->5701/tcp
hz3	hazelcast/hazelcast:5.3	"hz start"	hazelcast3	2 minutes ago	Up 2 minutes	0.0.0.0:5703->5701/tcp, [::]:5703->5701/tcp
ls1	apz_labs-logging-service-1	"uvicorn main:app --..."	logging-service-1	2 minutes ago	Up 2 minutes	8000/tcp
ls2	apz_labs-logging-service-2	"uvicorn main:app --..."	logging-service-2	2 minutes ago	Up 2 minutes	8000/tcp
ls3	apz_labs-logging-service-3	"uvicorn main:app --..."	logging-service-3	2 minutes ago	Up 2 minutes	8000/tcp

```
(.venv_APZ_general) julfy@julfy-Nitro-ANS15-57:~/Documents/6th_term/APZ/Task_3_micro_hazel/APZ_labs$
```

Console logs demonstrating the Hazelcast nodes finding each other and initializing the cluster connection

```
(.venv_APZ_general) julfy@julfy-Nitro-ANS15-57:~/Documents/6th_term/APZ/Task_3_micro_hazel/APZ_labs$ docker logs hz1 2>81 | grep -iE "members|cluster" | tail -8
2026-05-03 18:57:37,153 [ INFO] [hazelcast.internal.configoverride] Detected external configuration entries in environment variables: [hazelcast.clustername=dev]
2026-05-03 18:57:37,297 [ INFO] [hazelcast.system] [172.19.0.3]:5701 [dev] [5.3.8] Cluster name: dev
2026-05-03 18:57:40,360 [ INFO] [hazelcast.cluster.service] [172.19.0.3]:5701 [dev] [5.3.8]
2026-05-03 18:57:40,490 [ INFO] [hazelcast.fervent.cannon.thread.in-1] [hazelcast.tcpserverconnection] [172.19.0.3]:5701 [dev] [5.3.8] Initialized new cluster connection between /172.19.0.3:5701 and /172.19.0.4:44285
2026-05-03 18:57:40,537 [ INFO] [hazelcast.fervent.cannon.thread.in-2] [hazelcast.tcpserverconnection] [172.19.0.3]:5701 [dev] [5.3.8] Initialized new cluster connection between /172.19.0.3:5701 and /172.19.0.2:37087
2026-05-03 18:57:45,638 [ INFO] [hazelcast.fervent.cannon.priority-generic.operation.thread-0] [hazelcast.cluster.service] [172.19.0.3]:5701 [dev] [5.3.8]
2026-05-03 18:57:46,672 [ INFO] [hazelcast.fervent.cannon.priority-generic.operation.thread-0] [hazelcast.partition.statemanager] [172.19.0.3]:5701 [dev] [5.3.8] Initializing cluster partition table arrangement...
```

Curl POST and GET requests to /transaction being properly routed to different users (alice, bob, carol)

```
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" POST alice +100 "" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"alice", "amount":100}' ; echo
"" POST alice +100 ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" POST alice -30 "" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"alice", "amount":-30}' ; echo
"" POST alice -30 ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" POST bob +250 "" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"bob", "amount":250}' ; echo
"" POST bob +250 ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" POST bob -50 "" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"bob", "amount":-50}' ; echo
"" POST bob -50 ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" POST carol +500 "" ; curl -s -X POST http://localhost:8000/transaction -H 'Content-Type: application/json' -d '{"user_id":"carol", "amount":500}' ; echo
"" POST carol +500 ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" GET /accounts "" ; curl -s http://localhost:8000/accounts ; echo
"" GET /accounts ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" GET /user/alice "" ; curl -s http://localhost:8000/user/alice ; echo
"" GET /user/alice ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" GET /user/bob "" ; curl -s http://localhost:8000/user/bob ; echo
"" GET /user/bob ""

(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$ echo "" GET /tinings "" ; curl -s http://localhost:8000/tinings | python3 -m json.tool
Expecting value: line 1 column 1 (char 0)
(.venv_APZ_general) julfy@julyfy-Nitro-ANS15-S7:~/Documents/sth_tern/APZ/Task_3_micro_hazel/APZ_labs$
```

```
julfy@julyfy-Nitro-ANS15-S7:~$ docker logs -f ls1 &
[1] 36315
julfy@julyfy-Nitro-ANS15-S7:~$ INFO: Started server process [1]
INFO: Waiting for application startup.
[ls1] connected to hazelcast nodes=['hz1:5701', 'hz2:5701', 'hz3:5701']
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
docker logs -f ls2 &
[2] 36400
julfy@julyfy-Nitro-ANS15-S7:~$ INFO: Started server process [1]
INFO: Waiting for application startup.
[ls2] connected to hazelcast nodes=['hz1:5701', 'hz2:5701', 'hz3:5701']
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
docker logs -f ls3 &
[3] 36457
julfy@julyfy-Nitro-ANS15-S7:~$ INFO: Started server process [1]
INFO: Waiting for application startup.
[ls3] connected to hazelcast nodes=['hz1:5701', 'hz2:5701', 'hz3:5701']
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
[ls1] stored tid=1777835611519685-63df1ab1 user=alice amount=+1.0
INFO: 172.19.0.18:43240 - "POST /log HTTP/1.1" 200 OK
[ls1] stored tid=1777835611619283-8d137ef1 user=alice amount=+1.0
INFO: 172.19.0.18:43240 - "POST /log HTTP/1.1" 200 OK
[ls3] stored tid=1777835611638522-dd400c7d user=alice amount=+1.0
INFO: 172.19.0.18:48282 - "POST /log HTTP/1.1" 200 OK
[ls1] stored tid=1777835611649301-e257e266 user=alice amount=+1.0
INFO: 172.19.0.18:43240 - "POST /log HTTP/1.1" 200 OK
[ls2] stored tid=1777835611659799-6efae5a3 user=alice amount=+1.0
INFO: 172.19.0.18:608350 - "POST /log HTTP/1.1" 200 OK
[ls2] stored tid=1777835611673072-1a3cbe user=alice amount=+1.0
INFO: 172.19.0.18:608350 - "POST /log HTTP/1.1" 200 OK
[ls1] stored tid=1777835611723813-eacc067f user=alice amount=+1.0
INFO: 172.19.0.18:43240 - "POST /log HTTP/1.1" 200 OK
[ls1] stored tid=1777835611734308-e7d5efef user=alice amount=+1.0
INFO: 172.19.0.18:43240 - "POST /log HTTP/1.1" 200 OK
[ls3] stored tid=1777835611745849-941981a7 user=alice amount=+1.0
INFO: 172.19.0.18:48282 - "POST /log HTTP/1.1" 200 OK
[ls3] stored tid=1777835611755646-dede4550 user=alice amount=+1.0
INFO: 172.19.0.18:48282 - "POST /log HTTP/1.1" 200 OK
julfy@julyfy-Nitro-ANS15-S7:~$
```

```

INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
julfy@julfy-Nitro-AN515-57:~$ docker logs --tail 30 ls2
INFO:      Started server process [1]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
julfy@julfy-Nitro-AN515-57:~$ for i in $(seq 1 10); do
    curl -s -X POST http://localhost:8000/transaction \
        -H 'Content-Type: application/json' \
        -d '{"user_id":"alice","amount":1}' ; echo
done
{"transaction_id":"1777835611519685-63df1ab1","balance":1.0}
{"transaction_id":"1777835611619283-84137ef1","balance":2.0}
{"transaction_id":"1777835611630522-dd400c7d","balance":3.0}
{"transaction_id":"1777835611649381-e257e266","balance":4.0}
{"transaction_id":"1777835611659799-6efae5a3","balance":5.0}
{"transaction_id":"1777835611673072-13a3c1be","balance":6.0}
{"transaction_id":"1777835611723013-eaccc6f7","balance":7.0}
{"transaction_id":"1777835611734300-e7a5cfef","balance":8.0}
{"transaction_id":"1777835611745049-941901a7","balance":9.0}
{"transaction_id":"1777835611755646-6e4e4558","balance":10.0}
julfy@julfy-Nitro-AN515-57:~$

```

A bash for loop executing 10 POST requests to the facade-service for the user "alice" with an amount of 1. The terminal output shows the JSON responses, confirming the system generated unique transaction IDs and incremented the balance sequentially from 1.0 to 10.0

Logs from the counter-service confirming it is processing /balance POST requests. Directly below this, a psql command executes a SELECT query on the accounts table inside the PostgreSQL database, proving that the user "alice" has an accurately saved balance of 10

```
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
julfy@julfy-Nitro-ANS15-57:~$ for i in $(seq 1 10); do
  curl -s -X POST http://localhost:8000/transaction \
    -H 'Content-Type: application/json' \
    -d '{"user_id": "alice", "amount": 1}' ; echo
done
{"transaction_id": "1777835611519685-63df1ab1", "balance": 1.0}
{"transaction_id": "1777835611619283-84137ef1", "balance": 2.0}
{"transaction_id": "1777835611630522-dd400c7d", "balance": 3.0}
{"transaction_id": "1777835611649381-e257e266", "balance": 4.0}
{"transaction_id": "1777835611659799-6efae5a3", "balance": 5.0}
{"transaction_id": "1777835611673072-13a3c1be", "balance": 6.0}
{"transaction_id": "1777835611723013-eaccc6f7", "balance": 7.0}
{"transaction_id": "1777835611734300-e7a5cfef", "balance": 8.0}
{"transaction_id": "1777835611745049-941901a7", "balance": 9.0}
{"transaction_id": "1777835611755646-6e4e4558", "balance": 10.0}
julfy@julfy-Nitro-ANS15-57:~$ docker logs --tail 20 cs
INFO:      Started server process [1]
INFO:      Waiting for application startup.
[counter] connected to postgres, accounts table ready
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
INFO:      172.19.0.10:57982 - "POST /balance HTTP/1.1" 200 OK
julfy@julfy-Nitro-ANS15-57:~$ \echo "----"
----
julfy@julfy-Nitro-ANS15-57:~$ docker exec db psql -U user -d messages_db -c "SELECT user_id, balance FROM accounts ORDER BY user_id;"
 user_id | balance
-----+-----
  alice  |      10
(1 row)

julfy@julfy-Nitro-ANS15-57:~$
```

Terminal commands showing docker stop ls1 ls2 to simulate a dual-node failure. Subsequent curl commands prove that POST and GET requests for a "failover-test" user still succeed and return a valid balance during the outage

```
julfy@julfy-Nitro-ANS15-57:~$ echo "=== stop ls1 ls2 ==="; docker stop ls1 ls2
=== stop ls1 ls2 ===
ls1
ls2
```

```
julfy@julfy-Nitro-ANS15-57:~$ INFO:      Shutting down
INFO:      Shutting down
INFO:      Waiting for application shutdown.
INFO:      Application shutdown complete.
INFO:      Finished server process [1]
INFO:      Waiting for application shutdown.
INFO:      Application shutdown complete.
INFO:      Finished server process [1]
```

Terminal commands showing docker stop hz1 hz2 to simulate a distributed map failure. A subsequent POST request confirms that the remaining node successfully processed the transaction without data loss

```
julfy@julfy-Nitro-AN515-57:~$ echo "=== POST during outage ===" ; curl -s -X POST http://localhost:8080/transaction -H 'Content-Type: application/json' -d '{"user_id":"failover-test","amount":42}' ; echo
=== POST during outage ===
{"transaction_id":"1777835820351894-42fed69","balance":42.0}
julfy@julfy-Nitro-AN515-57:~$ echo "=== GET during outage ===" ; curl -s http://localhost:8080/user/failover-test ; echo
=== GET during outage ===
{"user_id":"failover-test","balance":42.0,"transactions":[{"transaction_id":"1777835820351894-42fed69","user_id":"failover-test","amount":42.0}]}
julfy@julfy-Nitro-AN515-57:~$ echo "=== /timings (note logging_failovers) ===" ; curl -s http://localhost:8080/timings | python3 -m json.tool
=== /timings (note logging_failovers) ===
{
  "logging_total_seconds": 0.22033960999815886,
  "counter_total_seconds": 0.06524753600024269,
  "logging_calls": 14,
  "counter_calls": 12,
  "logging_failovers": 2,
  "logging_avg_ms": 15.73854292843992,
  "counter_avg_ms": 5.437294666806891
}
```

```
INFO: Finished server process [1]
[ls3] stored tid=1777835820351894-42fed69 user=failover-test amount=+42.0
INFO: 172.19.0.10:45634 - "POST /log HTTP/1.1" 200 OK
INFO: 172.19.0.10:32800 - "GET /log/user/failover-test HTTP/1.1" 200 OK
julfy@julfy-Nitro-AN515-57:~$
```

```
sudo snap install ht
julfy@julfy-Nitro-AN515-57:~$ echo "=== stop hz1 hz2 ==="; docker stop hz1 hz2
=== stop hz1 hz2 ===
hz1
hz2
```

```
julfy@julfy-Nitro-AN515-57:~$ hi
Command 'hi' not found, but can be installed with:
sudo snap install hi
[1] Done docker logs -f ls1
[2]- Done docker logs -f ls2
julfy@julfy-Nitro-AN515-57:~$
```

```
sudo snap install hi
julfy@julfy-Nitro-AN515-57:~$ echo "=== stop hz1 hz2 ==="; docker stop hz1 hz2
=== stop hz1 hz2 ===
hz1
hz2
julfy@julfy-Nitro-AN515-57:~$ sleep 3
julfy@julfy-Nitro-AN515-57:~$ echo "=== POST after HZ partial outage ==="; curl -s -X POST http://localhost:8008/transaction -H 'Content-Type: application/json' -d '{"user_id":"hz-fallover","amount":7}' ; echo
=== POST after HZ partial outage ===
{"transaction_id":"1777836079452868-890de024","balance":7.0}
julfy@julfy-Nitro-AN515-57:~$
```

```
julfy@julfy-Nitro-AN515-57:~$ [ls3] stored tid=1777836079452868-890de024 user=hz-fallover amount=7.0
INFO: 172.19.0.10:39868 - "POST /log HTTP/1.1" 200 OK
```

```
julfy@julfy-Nitro-AN515-57:~$ echo "=== restart hz1 hz2 ==="; docker start hz1 hz2
=== restart hz1 hz2 ===
hz1
hz2
```

The terminal output of the perf_test.py script running 100,000 transactions. It displays the final wall time, throughput rates, and average latency metrics for two different load patterns: distinct user accounts and a single shared account

```
(.venv_APZ_general) julfy@julfy-Mitro-ANS15-S7:~/Documents/6th_term/APZ/Task_3_micro_hazel/AP2_labs$ python3 perf_test.py --scenario A --clients 10 --txns 10000

=== Scenario A: distinct accounts ===
clients=10, txns/client=10000, total=100000
client 7 (runA-user7): 2500/10000
client 1 (runA-user1): 2500/10000
client 2 (runA-user2): 2500/10000
client 8 (runA-user8): 2500/10000
client 0 (runA-user0): 2500/10000
client 9 (runA-user9): 2500/10000
client 5 (runA-user5): 2500/10000
client 4 (runA-user4): 2500/10000
client 3 (runA-user3): 2500/10000
client 6 (runA-user6): 2500/10000
client 7 (runA-user7): 5000/10000
client 0 (runA-user0): 5000/10000
client 8 (runA-user8): 5000/10000
client 1 (runA-user1): 5000/10000
client 2 (runA-user2): 5000/10000
client 4 (runA-user4): 5000/10000
client 6 (runA-user6): 5000/10000
client 3 (runA-user3): 5000/10000
client 5 (runA-user5): 5000/10000
client 9 (runA-user9): 5000/10000
client 4 (runA-user4): 7500/10000
client 7 (runA-user7): 7500/10000
client 0 (runA-user0): 7500/10000
client 6 (runA-user6): 7500/10000
client 8 (runA-user8): 7500/10000
client 1 (runA-user1): 7500/10000
client 5 (runA-user5): 7500/10000
client 2 (runA-user2): 7500/10000
client 3 (runA-user3): 7500/10000
client 9 (runA-user9): 7500/10000
client 4 (runA-user4): 10000/10000
client 0 (runA-user0): 10000/10000
client 7 (runA-user7): 10000/10000
client 1 (runA-user1): 10000/10000
client 8 (runA-user8): 10000/10000
client 5 (runA-user5): 10000/10000
client 6 (runA-user6): 10000/10000
client 2 (runA-user2): 10000/10000
client 3 (runA-user3): 10000/10000
client 9 (runA-user9): 10000/10000
wall time = 428.12 s
successful txns = 100000
failed txns = 0
throughput = 233.6 req/s
logging cumulative = 1895.15 s (avg 18.95 ms over 100000 calls, failovers=0)
counter cumulative = 2968.05 s (avg 29.68 ms over 100000 calls)
final balances = {'runA-user7': 10000.0, 'runA-user8': 10000.0, 'runA-user5': 10000.0, 'runA-user6': 10000.0, 'runA-user1': 10000.0, 'runA-user4': 10000.0, 'runA-user2': 10000.0, 'runA-user3': 10000.0, 'runA-user0': 10000.0, 'runA-user9': 10000.0}
ser9: 10000.0
(.venv_APZ_general) julfy@julfy-Mitro-ANS15-S7:~/Documents/6th_term/APZ/Task_3_micro_hazel/AP2_labs$
```

```
(.venv_APZ_general) julfy@julfy-Mitro-ANS15-S7:~/Documents/6th_term/APZ/Task_3_micro_hazel/AP2_labs$ python3 perf_test.py --scenario B --clients 10 --txns 10000

=== Scenario B: shared account ===
clients=10, txns/client=10000, total=100000
client 7 (runB-shared): 2500/10000
client 9 (runB-shared): 2500/10000
client 4 (runB-shared): 2500/10000
client 8 (runB-shared): 2500/10000
client 2 (runB-shared): 2500/10000
client 1 (runB-shared): 2500/10000
client 6 (runB-shared): 2500/10000
client 0 (runB-shared): 2500/10000
client 3 (runB-shared): 2500/10000
client 5 (runB-shared): 2500/10000
client 8 (runB-shared): 5000/10000
client 7 (runB-shared): 5000/10000
client 2 (runB-shared): 5000/10000
client 1 (runB-shared): 5000/10000
client 9 (runB-shared): 5000/10000
client 4 (runB-shared): 5000/10000
client 3 (runB-shared): 5000/10000
client 6 (runB-shared): 5000/10000
client 5 (runB-shared): 5000/10000
client 0 (runB-shared): 5000/10000
client 2 (runB-shared): 7500/10000
client 4 (runB-shared): 7500/10000
client 1 (runB-shared): 7500/10000
client 8 (runB-shared): 7500/10000
client 7 (runB-shared): 7500/10000
client 9 (runB-shared): 7500/10000
client 3 (runB-shared): 7500/10000
client 6 (runB-shared): 7500/10000
client 0 (runB-shared): 7500/10000
client 5 (runB-shared): 7500/10000
client 4 (runB-shared): 10000/10000
client 2 (runB-shared): 10000/10000
client 7 (runB-shared): 10000/10000
client 1 (runB-shared): 10000/10000
client 8 (runB-shared): 10000/10000
client 3 (runB-shared): 10000/10000
client 9 (runB-shared): 10000/10000
client 6 (runB-shared): 10000/10000
client 5 (runB-shared): 10000/10000
client 0 (runB-shared): 10000/10000
wall time = 429.20 s
successful txns = 100000
failed txns = 0
throughput = 233.0 req/s
logging cumulative = 1996.67 s (avg 19.07 ms over 100000 calls, failovers=0)
counter cumulative = 2967.02 s (avg 29.67 ms over 100000 calls)
final balances = {'runB-shared': 100000.0}
(.venv_APZ_general) julfy@julfy-Mitro-ANS15-S7:~/Documents/6th_term/APZ/Task_3_micro_hazel/AP2_labs$
```

```

{
  "balances": 1000000,
  "total_shared": 1000000,
  "total_unshared": 0
}
(.venv_APZ_general) julfy@julfy-Nitro-AN515-57:~/Documents/6th_tern/APZ/Task_3_micro_hazel/APZ_labs$ curl -s http://localhost:8000/timings | python3 -m json.tool
{
  "logging_total_seconds": 1906.6744228049029,
  "counter_total_seconds": 2967.023573999819,
  "logging_calls": 100000,
  "counter_calls": 100001,
  "logging_failovers": 0,
  "logging_avg_ms": 19.06674422804903,
  "counter_avg_ms": 29.669939040607783
}
(.venv_APZ_general) julfy@julfy-Nitro-AN515-57:~/Documents/6th_tern/APZ/Task_3_micro_hazel/APZ_labs$

```

A detailed comparison matrix directly contrasting the metrics from Lab 1 against Lab 3. The table breaks down wall time, throughput, and average operational times per call, including the percentage differences between the two architectural approaches

Metric	Lab 1 (in-memory, 1 ls)	Lab 3 (Hazelcast 3-node + PostgreSQL, 3 ls)	Δ
Scenario A — distinct accounts			
Wall time	272.72 s	428.12 s	+57.0 %
Throughput	366.7 req/s	233.6 req/s	-36.3 %
Logging avg / call	16.59 ms	18.95 ms	+14.2 %
Counter avg / call	15.98 ms	29.60 ms	+85.2 %
Logging cumulative	1658.53 s	1895.15 s	+14.3 %
Counter cumulative	1598.17 s	2960.05 s	+85.2 %
Failed txns	0	0	—
Logging failovers	n/a	0	—
Balances correct?	all 10 000	all 10 000	—
Scenario B — shared account			
Wall time	282.94 s	429.20 s	+51.7 %
Throughput	353.4 req/s	233.0 req/s	-34.1 %
Logging avg / call	17.08 ms	19.07 ms	+11.7 %
Counter avg / call	16.59 ms	29.67 ms	+78.8 %

Metric	Lab 1 (in-memory, 1 ls)	Lab 3 (Hazelcast 3-node + PostgreSQL, 3 ls)	Δ
Logging cumulative	1708.23 s	1906.67 s	+11.6 %
Counter cumulative	1658.52 s	2967.02 s	+78.9 %
Failed txns	0	0	—
Logging failovers	n/a	0	—
Balances correct?	shared = 100 000	shared = 100 000	—

Code: https://github.com/Bohdanok/APZ_labs/tree/micro_hazelcast